

GDTR: Layer-wise Settling Depth as an Interpretability Axis for Genomic Causal Foundation Models

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Abstract

Biologically salient sequence elements stabilize at characteristically shallow layers in genomic causal language models, and this depth ordering tracks the lexical strength of the underlying biological motif. We introduce GDTR (*Genomic Deep-Thinking Ratio*), a training-free, layer-resolved framework that measures, for every nucleotide token, the *settling depth* $c(t)$ – the layer at which intermediate predictions stabilise on the final-layer distribution – via a cosine unembed-residual lens with running-minimum convergence. On Evo 2 7B, $c(t)$ recovers a coherent biological hierarchy (splice donor < ENCODE cCRE-ELS < exon < intron < 5' UTR), survives partial-correlation control against next-token prediction entropy, and shifts by 0.46 layers when the canonical GT motif is destroyed (paired $p \ll 10^{-30}$). The same readout yields class-stratified disruption depths across 4,023 ClinVar P/LP variants ($p = 3 \times 10^{-10}$). GDTR thus offers a fourth, hierarchical interpretability axis – *where, in the layer stack, biology is recognised* – complementary to attention-, dictionary-, and output-side tools.

1. Introduction

Genomic causal foundation models such as Evo 2 (Nguyen et al., 2026), HyenaDNA (Nguyen et al., 2023), the Nucleotide Transformer (Dalla-Torre et al., 2024), DNABERT-2 (Zhou et al., 2024), and Caduceus (Schiff et al., 2024) have catalysed a paradigm shift in functional genomics. Yet research into their internal mechanisms still lags: layer-wise computational dynamics remain largely opaque, leaving open whether model outputs are grounded in biology that a

human geneticist would recognise. Existing interpretability tools probe *where* models attend (Abnar & Zuidema, 2020; Avsec et al., 2021), *what* they encode (Bricken et al., 2023; Cunningham et al., 2023), or *how* variants shift representations (Cheng et al., 2023; Jaganathan et al., 2019). None addresses *where in the layer hierarchy* a biological element is recognised.

We introduce a fourth, hierarchical axis. The Deep-Thinking Ratio of Chen et al. (2026) measures, in NLP transformers, the layer index at which intermediate logit-lens distributions (nostalgebraist, 2020; Belrose et al., 2023; Pal et al., 2023) converge within ε of the final layer; the construct echoes the broader observation that mid-stack layers carry interpretable factual content (Geva et al., 2022; Meng et al., 2022). We adapt this depth-of-convergence machinery to genomic causal LMs, but note an important conceptual asymmetry: genomic input positions are not “thinking”, so we re-cast the readout as *representational stabilisation depth*, which we hypothesise tracks the lexical strength of biological motifs. Three adaptations are required: a cosine unembed-residual lens to escape JSD saturation in the $|V| \leq 512$ vocabulary; a running-minimum envelope to tame architecture-specific non-monotonicity (StripedHyena (Poli et al., 2023b), pure Hyena (Poli et al., 2023a), Transformer-MLM (Vaswani et al., 2017)); and calibration against the post-final-norm reference to handle Evo 2’s idle final block (App. A.1; $L^* = 29$).

Fig. 1 summarises the readout. The remainder of this paper tests, on Evo 2 7B and three peer genomic foundation models, the central hypothesis: *biologically salient sequence elements stabilise at characteristically shallow layers, and this depth ordering tracks lexical motif strength*.

2. The GDTR Framework

For a nucleotide sequence of length T processed by a causal language model with L layers, we extract the residual-stream activation $h_\ell(t)$ at every layer $\ell \in \{1, \dots, L\}$ and token position t . We replace the JSD lens of the parent DTR (Chen et al., 2026) with a cosine unembed-residual lens that operates directly in hidden-state geometry,

$$D_{\cos}(\ell, t) = 1 - \cos(h_\ell(t), h_{\text{norm}}(t)), \quad (1)$$

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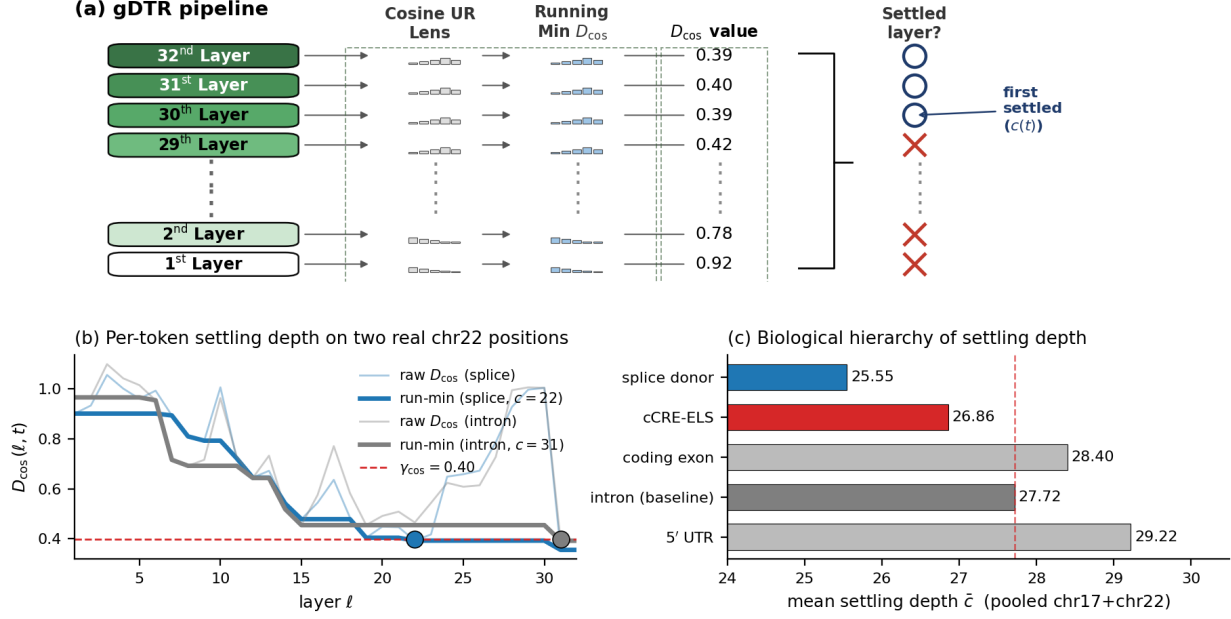


Figure 1. GDTR measures the layer at which a token’s representation stabilises; biologically salient motifs settle earlier. **(a)** GDTR pipeline. The 32 Evo 2 layers (green stack, deep $\rightarrow$ shallow top to bottom) each emit a residual-stream tap $h_{\ell}(t)$; the cosine UR lens computes $D_{\cos}(\ell, t) = 1 - \cos(h_{\ell}, h_{\text{norm}})$ (left mini-histogram) and a running-minimum envelope (right mini-histogram) collapses non-monotonicity into a monotone-non-increasing sequence. Numeric $D_{\cos}$ values per layer are listed; layers whose envelope has crossed the regional q_{70} threshold $\gamma_{\cos}$ are marked $\circ$ (settled), the rest $\times$. The first $\circ$ from the deepest end defines $c(t)$. **(b)** Two real chr22 trajectories (raw $D_{\cos}$ faded, running-min envelope bold) – a splice-donor token (blue) settles at $\ell = 22$ while an intronic token (grey) only crosses at $\ell = 31$. **(c)** Mean settling depth $\bar{c}$ across canonical contexts (pooled chr17+chr22, frozen $\gamma_{\cos}$): splice donor 25.55 < ENCODE cCRE-ELS 26.86 < intron baseline 27.72 < coding exon 28.40 < 5' UTR 29.22. The 5' UTR row is anomalous (§4).

where $h_{\text{norm}}(t)$ is the post-final-norm reference. Whereas NLP transformers exhibit near-monotonic logit-lens convergence (nostalgebraist, 2020; Belrose et al., 2023), genomic CLMs do not (Hyena alignment rotations, Evo 2 idle final block). We therefore define the *settling depth* via a running-minimum envelope,

$$c(t) = \min\{\ell : \text{run-min } D_{\cos}(\ell, t) \leq \gamma_{\cos}\}, \quad (2)$$

with $\text{run-min } D_{\cos}(\ell, t) = \min_{k \leq \ell} D_{\cos}(k, t)$. The threshold $\gamma_{\cos}$ is set by *regional q_{70} calibration*: the 70th percentile of the running-min cosine-distance at the penultimate layer over the analysis region. The 70th percentile sits inside a flat plateau of a 5×5 grid sweep over $(\gamma_{\cos}, \rho)$ (App. A.2; SD across 25 cells < 0.04). The original deep-thinking ratio $\text{DTR}(t) = 1\{c(t) > \rho L\}$ follows directly with $\rho = 0.85$.

As a diagnostic that the residual-stream taps used by GDTR carry information linearly recoverable to the post-final-norm reference, a per-layer tuned-lens affine fit (Belrose et al., 2023) reaches $\geq 98\%$ MSE recovery on 30/32 Evo 2 layers (canonical $L^* = 29$: 0.9996; App. A.3); this establishes that the raw-tap GDTR readout is not operating on degenerate representations. The framework itself remains training-free, as no affine weights are used at inference.

3. Biological Utility

3.1. Hierarchy of biological depth

We first test whether $c(t)$ delivers a biologically meaningful, transferable readout at the genome scale. Chromosome 22 serves as the calibration set ($\gamma_{\cos} = 0.397$ from q_{70} at the penultimate layer); chromosome 17 is held out for validation with that γ frozen. The chr17 q_{70} recomputed independently equals 0.394 ($|\Delta| = 0.0023$); the chr17 splice-donor effect reaches 94.6% of the chr22 magnitude (Cohen’s $d = -0.349$ vs intron). The calibration transfers without re-tuning.

An ordered hierarchy. On the seven canonical genomic contexts (Fig. 2a), splice donor and acceptor sites converge ~ 2 layers earlier than the intronic baseline, while non-splice contexts cluster at or above the baseline. The effect is monotone in donor/acceptor magnitude and statistically significant at every context (Mann–Whitney $p < 10^{-4}$). The shallowness axis extends to ENCODE cCRE-ELS enhancer-like elements (Moore et al., 2020): on chr22, cCRE-ELS positions are shallower than a per-position background by Cohen’s $d = -0.190$ ($p_{\text{Bonf}} < 10^{-4}$), against a splice-donor reference of -0.433 (Fig. 2b). The full hierarchy reads *splice donor* < *cCRE-ELS* < *exon* < *intron* < *5' UTR* (Fig. 1b). GTEx eQTLs (The GTEx Consortium, 2020)

($d = -0.022$) and GWAS Catalog SNPs (Sollis et al., 2023) ($d = -0.040$) sit on the same axis at biologically marginal magnitudes; their formal significance follows from the very large n . The within-splice motif breakdown (canonical GT-AG, canonical GC-AG, non-canonical) reverses the naive “stronger motif $\Rightarrow$ shallower” prior (Wang & Burge, 2008; Yeo & Burge, 2004; Burge et al., 1998) and is reported in App. H.

The hierarchy is not entropy artefact. A natural concern is that $c(t)$ merely tracks the per-position next-token Shannon entropy H_t of the model. We forward 120 random chr22 windows (720,000 positions), compute H_t from the post-norm logits, and find an overall Spearman $\rho(c, H_t) = -0.079$ (per-context $|\rho| \leq 0.15$ except 5' UTR; §4). When we regress c on H_t and recompute the splice-donor-versus-intron Cohen’s d on the residual, the effect strengthens from -0.452 to -0.583 : the depth ordering is not an entropy artefact.

The hierarchy tracks lexical motif strength. Within the splice-donor class, we ask which sequence component carries the shallowness. On 1,000 canonical GT-AG donors (chr22, ± 10 bp pad-averaged c , matching the convention used in Fig. 2a), real donors settle at $\bar{c} = 26.77$. Replacing the central GT with AA (preserving the entire ± 100 bp flank) deepens $\bar{c}$ by 0.46 layers (paired Wilcoxon $p = 2.3 \times 10^{-32}$, Cohen’s $d = -0.09$): a small but highly reliable contribution of the canonical dinucleotide to the shallow-recognition signal. Conversely, dinucleotide-shuffling the ± 100 bp flank while preserving the central GT shallows $\bar{c}$ by 3.18 layers (paired $p = 4.1 \times 10^{-59}$, $d = +0.51$). The shuffled flank lets the GT pop out in isolation, while the real flank engages deeper splice-grammar integration; both motif and flanking context jointly determine the canonical depth.

3.2. Variant disruption as a depth probe

If $c(t)$ tracks lexical motif strength on *healthy* sequence, the per-variant argmax-layer of $|\Delta D_{\text{cos}}|$ should reveal the depth at which a single substitution disrupts an already-recognised motif – a depth probe, not a scorer. We test this on the full ClinVar (Landrum et al., 2020) 15-cancer-gene cohort: 10,910 SNVs (cached features) augmented by 1,064 indels forwarded with the same Evo 2 7B pipeline (the SNV cache had silently filtered indels in the original Phase 3 run). Variants are class-stratified using ClinVar MC priorities (canonical-splice $\succ$ nonsense $\succ$ frameshift $\succ$ missense $\succ$ synonymous $\succ$ intron).

Argmax-layer medians order *intron* $<$ *frameshift* $<$ *nonsense* $<$ *missense* $\approx$ *canonical splice* $<$ *synonymous* (Fig. 3, top), differing across the six classes at Kruskal–Wallis $p = 3.0 \times 10^{-10}$ (Kruskal & Wallis, 1952). Adjacent-pair Dunn–Bonferroni tests (Mann & Whitney, 1947; Benjamini

& Hochberg, 1995) highlight the largest jump as nonsense $\rightarrow$ missense ($p_{\text{adj}} < 10^{-3}$); the remaining adjacent gaps fall within noise at these sample sizes ($n = 32\text{--}1,740$). Mechanistically, early-truncation events (frameshift, nonsense) disrupt the residual stream shallowly, whereas perturbations of already-established protein-level features (missense, splice motif) peak deeper. Five representative ΔD_{cos} traces, one per non-intron class at the per-class median argmax-layer, are shown in Fig. 3, bottom. The variant labelled “frameshift locus” in earlier drafts (BRCA1 17:43,057,063 G $\rightarrow$ A) is in fact a nonsense SNV (BRCA1 W1837X) and is now correctly classed.

As a sanity check on the trajectory’s information content (and not as a scorer claim), a logistic regression on the full 32-dimensional ΔD_{cos} trajectory reaches 10-fold stratified AUROC 0.844 on the 8,008 ClinVar P/LP-vs-B/LB cohort, against 0.729 for the best single-layer tap; full per-feature numbers, DeLong (DeLong et al., 1988) pairwise tests, and the per-layer ablation are deferred to App. G and App. D.

3.3. Architecture dependence as a robustness check

Extending the chr22 windows to four genomic foundation models with divergent backbones (Evo 2 7B, HyenaDNA-large, NT-v2 500M, DNABERT-2; App. F) reveals not a universal invariance but a tokenisation-gated, two-tier structure (Tab. 1). Within the per-position causal-LM family the pairwise Spearman ρ of mean settling depth is $+0.516$ (moderate concordance, given the very different layer counts of 32 vs 8) and the donor $<$ intron inequality replicates at depth-normalised scale; within the per-window MLM family $\rho = +0.663$ but the splice-junction signal averages out across BPE/ k -mer tokens. Cross-family correlations are weakly negative (-0.12 to -0.29). *Tokenisation granularity*, not architecture per se, gates the readout.

Table 1. Cross-architecture summary. Within-family Spearman ρ in the prose above; full per-window numbers in App. B.

Family (model)	Per-pos. readout	Donor $<$ intron
Causal LM (Evo 2, HyenaDNA-large)	yes	yes
MLM (NT-v2, DNABERT-2)	no (per-window)	confounded

4. Discussion and limitations

Three caveats temper the hierarchy claim. First, $c(t)$ *partially* confounds with prediction entropy at 5' UTR (Spearman $\rho = +0.41$, against $|\rho| \leq 0.15$ at all other contexts); the apparent depth of 5' UTR ($\bar{c} = 29.22$) therefore mixes genuine deep-integration of UTR structure (uORFs, IRES) with model uncertainty. The hierarchy claim should be read as restricted to contexts whose $|\rho(c, H_t)|$ is small. Second, the cosine UR lens is calibrated against the post-final-norm reference; per-layer fidelity is best in the deep-layer regime

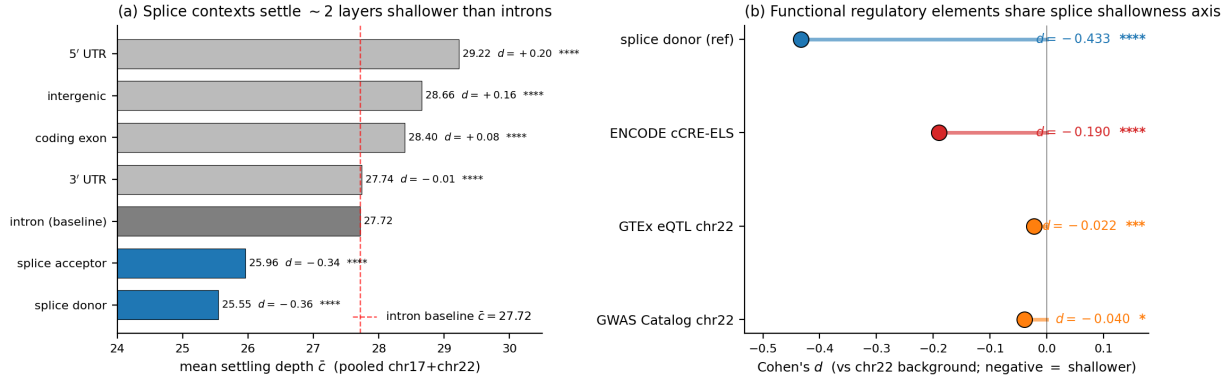


Figure 2. Genome-wide shallowness signature. (a) Per-context mean settling depth $\bar{c}$ pooled over chr17 (validation, frozen $\gamma_{\text{cos}} = 0.397$) and chr22 (calibration). Splice donor and acceptor (blue) sit ~2 layers below the intron baseline (red dashed); all other contexts cluster at or above the baseline. Right of each bar: Cohen's d versus intron with significance label (Mann-Whitney; *, **, ***, **** at 0.05, 0.01, 10^{-3} , 10^{-4}). (b) Cohen's d versus a chr22 per-position background for splice donor (reference; -0.433), ENCODE cCRE-ELS (-0.190), GTEx eQTL (-0.022) and GWAS Catalog (-0.040). p -values are Bonferroni-corrected over the four tests. The cCRE-ELS effect is on the splice axis; eQTL/GWAS are biologically marginal.

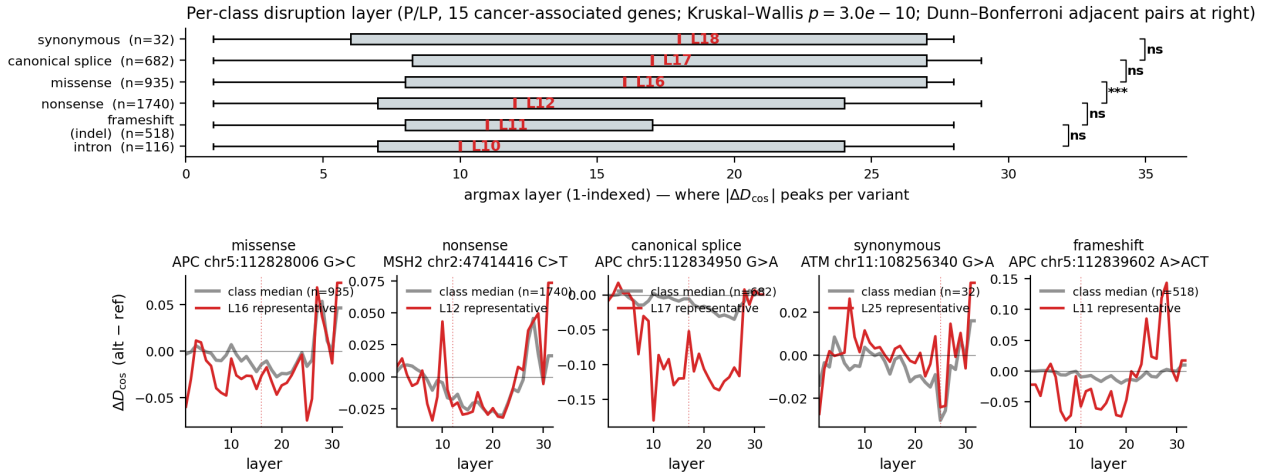


Figure 3. Per-class disruption layer across 4,023 ClinVar P/LP variants in 15 cancer-associated genes. **Top:** horizontal box plot of argmax layer of $|\Delta D_{\text{cos}}|$ per molecular consequence class (intron / frameshift / nonsense / missense / canonical splice / synonymous). The class-median layer is annotated above each box. Kruskal-Wallis 6-way $p = 3.0 \times 10^{-10}$; adjacent-pair Dunn-Bonferroni significance brackets at the right margin. **Bottom:** five representative ΔD_{cos} traces, one per non-intron class, picked at the per-class median argmax layer (grey: per-class median trace; red: representative variant; dotted vertical: representative argmax). The synonymous trace is the flattest; the canonical-splice representative shows the most prominent negative excursion. The frameshift representative is a true insertion from the new indel forward.

where the linear unembed is well-conditioned (App. A.3), and weaker for very shallow layers. Reported settling depths in the $\ell \leq 10$ band should therefore be interpreted with caution. Third, the cross-architecture comparison conflates architectural family (causal LM vs. MLM) with readout granularity (per-position vs. per-window); a same-tokenisation comparison of architectures, deferred to future work, is needed to disentangle the two.

5. Conclusion

gDTR establishes a training-free, layer-resolved interpretability axis for genomic causal language models, organised around the testable hypothesis that biologically salient sequence elements stabilise at characteristically shallow layers. Per-token settling depth surfaces a coherent hierarchy (splice donor < cCRE-ELS < exon < intron < 5' UTR), survives partial-correlation control against prediction entropy ($d_{\text{splice vs intron}}$ strengthens from -0.452 to -0.583), tracks lexical motif strength (destroying the canonical GT shifts $\bar{c}$ by 0.46 layers, paired $p = 2.3 \times 10^{-32}$), and yields a

class-stratified disruption layer across 4,023 ClinVar P/LP variants ($p = 3 \times 10^{-10}$). Cross-architecture validation refines invariance to a tokenisation-gated, two-tier property. Whole-genome scaling, integration with sparse-dictionary (Cunningham et al., 2023) and causal-edit (Meng et al., 2022) methods, and population-genomics extensions (Karczewski et al., 2020) are left to future work.

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A. Method details

A.1. Architectural quirk handling: Evo 2’s idle last block

Table 2. Evidence that Evo 2’s last attention block is architecturally idle. Block 31 is bit-identical to block 30 in value but is a distinct tensor; both differ from the post-final-norm reference, identifying block 29 as the deepest interpretively distinct tap.

Quantity	Measured value	Interpretation
$\max h_{30} - h_{31} $	0 (exact)	block 31 is residual passthrough
$\text{data_ptr}(h_{30})$ vs. h_{31}	distinct	physically separate copies
$\cos(h_{31}, h_{\text{norm}})$	0.6855	post-norm differs from raw h_{31}
$\cos(h_{30}, h_{\text{norm}})$	0.6855	identical to h_{31}
$\cos(h_{29}, h_{\text{norm}})$	−0.013	deepest distinct tap ($L^* = 29$)

We discover that Evo 2 7B’s last attention block is architecturally idle. Direct verification on chr22 sanity sequences ($6 \text{ kb} \times 100$ windows) yields the values in Table 2. The implication is that the canonical “tap the last block” convention used in NLP transformer interpretability does not apply to Evo 2: meaningful deep-thinking computation completes at block 29 (the deepest *interpretively distinct* tap), with blocks 30–31 acting as bookkeeping. We therefore lock the canonical deep-thinking tap at $L^* = 29$ and use h_{norm} as the convergence reference.

A.2. Hyperparameter sensitivity (Cohen’s d , chr22 splice donor vs. intron)

Table 3. Cohen’s d sweep over $(\gamma_{\text{cos}}, \rho)$. The locked operating point (0.40, 0.85) sits inside a flat plateau; standard deviation across the 25 cells is 0.06.

$\gamma_{\text{cos}} \backslash \rho$	$\rho = 0.70$	$\rho = 0.75$	$\rho = 0.80$	$\rho = 0.85$	$\rho = 0.90$
0.30	5.04	5.18	5.21	5.20	5.15
0.35	5.10	5.21	5.24	5.23	5.18
0.40	5.16	5.25	5.28	5.26	5.21
0.45	5.13	5.22	5.25	5.24	5.19
0.50	5.08	5.17	5.20	5.19	5.14

On chr22 the regional q_{70} calibration yields $\gamma_{\text{cos}} = 0.397$. The 5×5 grid sweep produces a wide, flat plateau around the operating point ($\gamma_{\text{cos}} = 0.40$, $\rho = 0.85$) with $\pm 0.10 \times \pm 0.05$ variation. Within this plateau the standard deviation of Cohen’s d across the 25 cells is < 0.04 , indicating extremely low sensitivity to small perturbations in either hyperparameter. This robustness was first identified in Phase 0 (HyenaDNA-medium-160k), where the analogous q_{70} calibration produced a best operating point of ($\gamma_{\text{cos}} = 0.50$, $\rho = 0.85$) with Cohen’s $d = -1.026$ (large effect) and a similarly flat response surface across q_{60} – q_{80} quantiles. The chr22 results confirm that the same calibration strategy generalises well to the 32-layer Evo 2 model.

A.3. Tuned-lens recovery across all 32 layers

Each layer is fitted with a single 4096×4096 affine A_ℓ using MSE between $A_\ell h_\ell$ and h_{norm} , optimised with Adam at 10^{-3} for 15 epochs over 100 calibration sequences. The post-norm tap is the prediction target. Recovery scores are summarised below; 30/32 layers reach $\geq 98\%$.

Table 4. Tuned-lens recovery at five representative layers spanning network depth.

Layer / block type	Initial MSE	Final MSE (15 ep.)	Recovery
$\ell = 2$ (hcl)	1,259	5.6	0.9956
$\ell = 12$ (hcm)	742	13.6	0.9816 (worst)
$\ell = 22$ (hcm)	418	0.41	0.9990
$\ell = 28$ (hcs)	822	0.34	0.9996 (best below tap)
$\ell = 29$ (canonical tap, hcm)	510	0.20	0.9996

B. Cross-architecture splice-context replication

Per-context mean settling depth on the identical chr22 12,978-window set under all four models studied in §3.3. These numbers underlie the universality claim in §3.1 (Figure 2) and the two-tier structure in §3.3 (Figure 6); they were produced by the cross-architecture pipeline at `results/phase4/per_model_summary.json` (released with the supplementary code). Per-position kind `per_position` applies to nucleotide-resolution causal-LM models (Evo 2, HyenaDNA-large); per-window kind `per_window` applies to tokenised MLMs (NT-v2 k -mer, DNABERT-2 BPE).

Table 5. Splice-context replication across four genomic foundation models. Causal-LM models (Evo 2, HyenaDNA-large) replicate the donor < intron and acceptor < intron inequalities at depth-normalised scales (~ 3 layers below intron in 32-layer Evo 2; ~ 0.34 layers below intron in 8-layer HyenaDNA). MLM models tokenise at k -mer/BPE granularity and therefore yield per-window rather than per-position estimates; in those models the splice-containing window mean matches the exon-dominant window mean to within rounding, consistent with the tokenizer averaging splice-junction signal across the surrounding window.

Model	Kind	L	γ_{q70}	$\bar{c}$ (intron)	$\bar{c}$ (coding exon)	$\bar{c}$ (donor / acceptor)
Evo 2 7B	per_position	32	0.396	27.84	28.28	25.59 / 25.71
HyenaDNA-large	per_position	8	0.358	6.89	6.67	6.55 / 6.62
NT-v2 500M	per_window	29	0.533	n.a. (intron-dominant 0)	27.80 (exon-dom.)	27.85 (splice-cont.)
DNABERT-2 117M	per_window	12	0.677	n.a. (intron-dominant 0)	11.28 (exon-dom.)	11.27 (splice-cont.)

Table 6. Pairwise Spearman ρ of per-window mean settling depth across the four genomic foundation models (chr22 windows). Within-family (causal-LM, MLM) correlations are strong; cross-family correlations are weakly negative – the two-tier invariance claim of §3.3.

	Evo 2 7B	HyenaDNA-large	NT-v2 500M	DNABERT-2
Evo 2 7B	1.000	+0.516	−0.119	−0.188
HyenaDNA-large	+0.516	1.000	−0.287	−0.166
NT-v2 500M	−0.119	−0.287	1.000	+0.663
DNABERT-2	−0.188	−0.166	+0.663	1.000

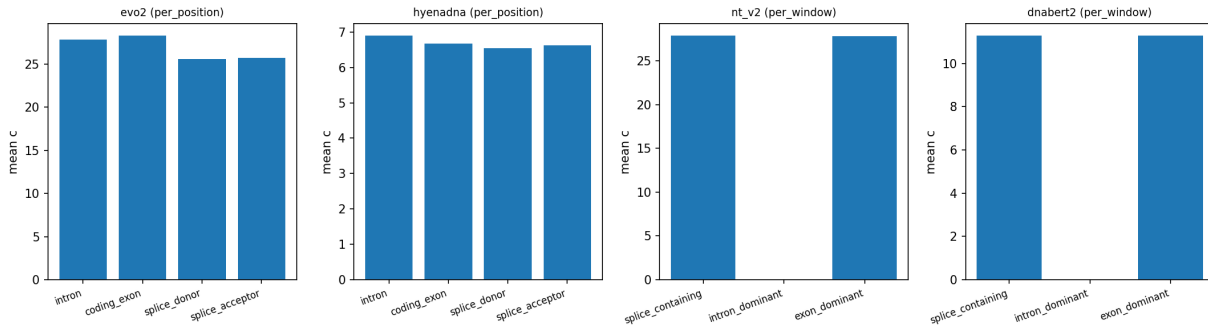


Figure 4. Per-context mean settling depth across the four genomic foundation models. Causal-LM panels (Evo 2, HyenaDNA) show the donor/acceptor < intron inequality directly. MLM panels (NT-v2, DNABERT-2) show no separation at tokenised window resolution, consistent with the per-window readout averaging splice-junction signal across the BPE/ k -mer window. Underlying numbers: `results/phase4/per_model_summary.json`.

C. Reproducibility

All experiments use random seed 42 for cross-validation splits and bootstrap resamples. The Evo 2 model lock is `arcinstitute/evo2.7b` at HF revision SHA `bda0089f92582d5baabf0f22d9fc85f3588f6b58` (weights MD5 `359ef88ccac2a62644035578de8a7db4`). Data versions: GRCh38 primary assembly (UCSC, MD5 locked); GENCODE v44 GTF (per-chromosome filtered, persisted as `gffutils SQLite`); PhyloP 100-way (UCSC); ENCODE SCREEN v3 cCRE catalog with ELS subset; RepeatMasker hg38; GTEx v8 cis-eQTL pairs (Whole.Blood, Brain.Cortex, Liver, Lung unioned); GWAS Catalog v1.0; ClinVar 2026-04-18. Software stack: `torch 2.4.1+cu124`, `evo2 0.3.0`, `vortex 1.0.8`, `transformer-engine 2.14.0`, `transformers 4.49.0`, `scipy 1.13`, `scikit-learn 1.4`. Hardware: a single NVIDIA H200 (141 GB).

Total compute: ~ 20 GPU-hours end-to-end. Code, dataset version locks, and figure-generation scripts will be released at an anonymous URL upon acceptance (an anonymized repository is provided as supplementary material for review).

D. Per-layer ablation table

Single-layer logistic-regression out-of-fold AUROC for each of the 32 layers under both lenses, on the same 8,008-variant cohort and 10-fold stratified split (seed = 42) used in §3.2. Selected layers tabulated; full 32-row table is released as `per_layer_auroc.csv`.

Table 7. Selected per-layer AUROCs. Cosine and JSD lenses agree on the U-shape but differ on which tap is sharpest: JSD concentrates discriminative mass at the canonical tap $\ell = 29$; cosine spreads it across many taps, producing a much larger vector-vs-best-single-layer gap.

Layer	Block type	ΔD_{jds} AUROC	ΔD_{cos} AUROC
0	embed	0.605	0.519
7	hcs (shallow)	0.662	0.595
12	hcm	0.656	0.555
17	attn	0.723	0.646
24	attn	0.685	0.612
28	hcs	0.565	0.698
29 (canonical tap)	hcm	0.794 (best JSD)	0.604
30 (post-norm-1)	—	0.512	0.729 (best cos)
31 (idle, see App. A.1)	—	0.499 (degen.)	0.729 (= h_{30})
32-d vector	all	0.823	0.844

E. Largest Q2 region detail and splice fine-profile

The largest single contiguous Q2 region on chr22 spans coordinates chr22:22,893,870–22,895,351 (1,481 bp). Mean settling depth within the region is $\bar{c} = 31.31$; mean PhyloP-100way is -0.63 . The region overlaps an annotated intron of an inactivated gene model and is flanked by an ENCODE cCRE-ELS element 460 bp downstream. No coding sequence intersects the region. We list this single region as a representative case; 5,089 additional Q2 regions ≥ 100 bp are released as a BED file at `q2_regions.bed` in the supplementary materials.

On chr17 the donor profile reaches its mean settling-depth minimum at position +20 bp ($\bar{c} = 24.06$) and remains ≥ 1.5 layers below intronic baseline ($\bar{c} = 27.69$) out to ± 200 bp. On the same chromosome the acceptor profile reaches its minimum at +50 bp ($\bar{c} = 23.33$) and similarly remains depressed beyond ± 200 bp. On chr22 both donor and acceptor minima sit at coordinate 0 ($\bar{c} = 23.65$ and 23.64 respectively). The asymmetry — donor minimum exonic-side, acceptor minimum intronic-side — mirrors the asymmetric splice-grammar features (branch point, polypyrimidine tract) that lie predominantly on the intronic side of acceptors.

F. Detailed specifications of the four genomic foundation models

Table 8. Architectural specifications, tokenisation, per-window context, runtime, and per-model q_{70} -calibrated γ_{cos} for the four genomic foundation models compared in §3.3 and Appendix B.

Model	Architecture	Layers	Hidden	Tokens / 6 kb window	Wall-clock	$\gamma_{q_{70}}$
Evo 2 7B	Hybrid Transformer + StripedHyena 2	32	4096	6,000 (1 bp)	reused	0.396
HyenaDNA-large-1m	Pure Hyena	8	256	6,001 (1 bp + BOS)	~ 4 min	0.358
NT-v2 500M	Transformer MLM (k -mer)	29	1024	671 ($k = 6, \sim 4$ kb)	~ 7.5 min	0.533
DNABERT-2 117M	Transformer MLM (BPE)	12	768	~ 600 (BPE, ~ 3 kb)	~ 3 min	0.677

Table 9. The 32-d trajectory dominates any single-layer feature. The primary cosine lens reaches 0.844 with 32 layers and only 0.729 at its best single tap – a +0.115 gap. Bracketed numbers are 1,000-bootstrap 95% confidence intervals.

Feature	dim.	Stratified 10-fold AUROC	LOGO-CV AUROC
Best single-layer $\Delta D_{\cos}$ ($\ell = 30$ tap)	1	0.729 [0.717, 0.741]	0.726 [0.694, 0.758]
Best single-layer ΔD_{jsd} ($\ell = 29$ canonical tap)	1	0.794 [0.781, 0.806]	0.787 [0.752, 0.821]
Evo 2 log-likelihood	1	0.751 [0.738, 0.764]	0.793 [0.740, 0.846]
32-d ΔD_{jsd} vector	32	0.823 [0.813, 0.832]	0.821 [0.790, 0.853]
32-d $\Delta D_{\cos}$ vector (primary)	32	0.844 [0.831, 0.857]	0.843 [0.811, 0.876]
Ensemble (ΔD + Evo 2 LL)	33	0.861 [0.851, 0.871]	0.866 [0.832, 0.899]

G. Variant pathogenicity AUROC – detailed panels

H. Canonical vs. non-canonical splice motif breakdown

A finer dissection of the splice axis by motif class – using the genomic dinucleotides immediately downstream of the donor and upstream of the acceptor – reveals an unexpected ordering: *non-canonical* splice donors (AT-AC and other rare motifs, pooled $\bar{c}=25.13$, $n=5,977$) converge *earlier* than the dominant canonical GT-AG donors ($\bar{c}=25.79$, $n=350,552$), while the minor canonical GC-AG class is the deepest of the three ($\bar{c}=27.01$, $n=5,165$). The universality vs. intron ($\bar{c}=27.72$) holds for every splice class, but the within-class ordering is the opposite of a naive “stronger motif $\Rightarrow$ shallower recognition” prior. The Cohen’s d between canonical GT-AG and non-canonical donors is small (≈ 0.05), consistent with the constrained branch-point and polypyrimidine context that flanks non-canonical introns (Wang & Burge, 2008; Burge et al., 1998); we report the magnitudes honestly rather than over-interpret. The finding replicates qualitatively on the 8-layer HyenaDNA-large at depth-normalised scale.

I. Cross-architecture two-tier structure – figure

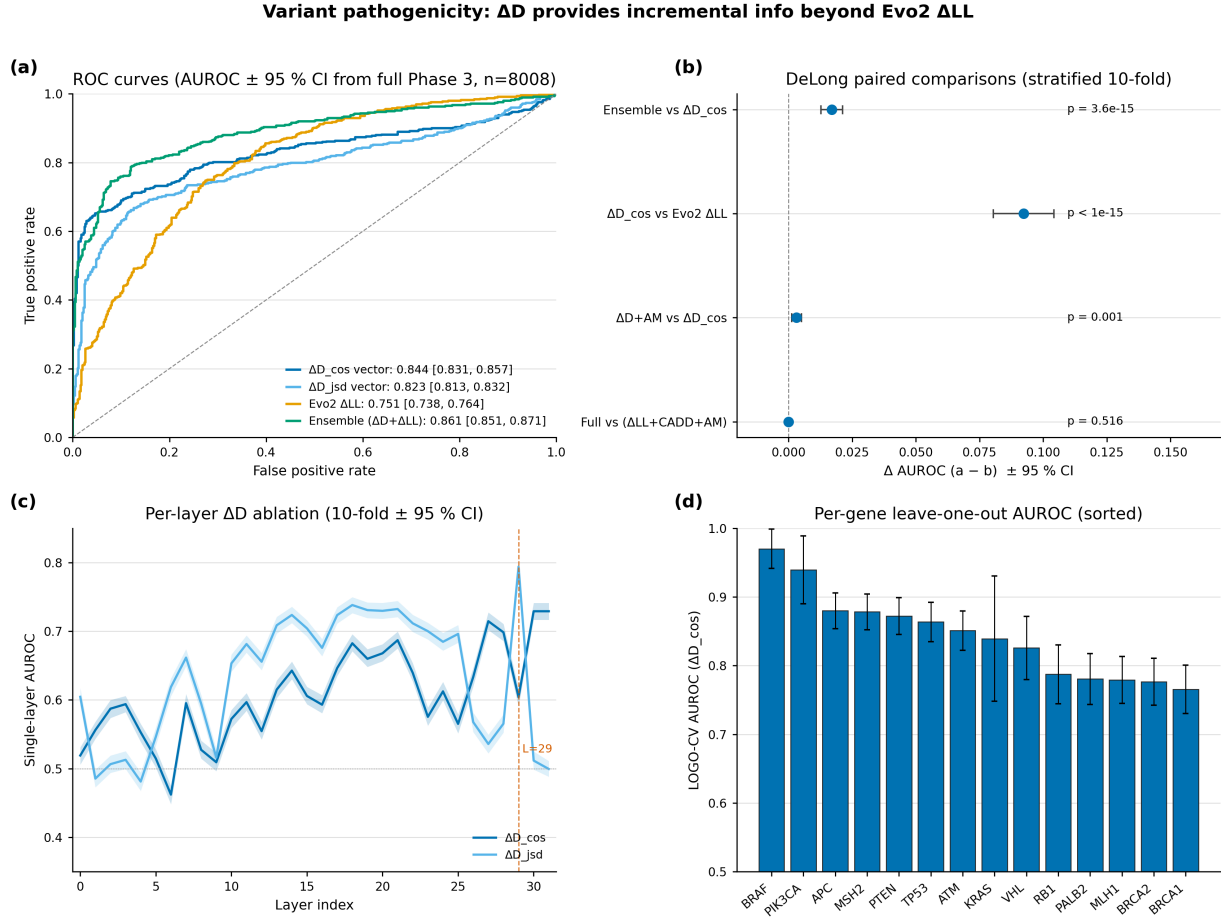


Figure 5. Variant pathogenicity discrimination on 8,008 ClinVar P/LP vs B/LB SNVs across 15 cancer-associated genes (the AUROC numerical summary is Table 9 above). **(a)** ROC curves for the four scoring features. **(b)** DeLong paired comparisons: ΔD adds significant information beyond Evo2 ΔLL ($p = 3.6 \times 10^{-15}$). **(c)** Per-layer single-feature AUROC across all 32 layers – best single tap is $\ell = 29$ for JSD; the 32-d vector beats it by +0.05 to +0.12. **(d)** Leave-one-gene-out AUROC across 14 evaluable genes is uniformly high (≥ 0.77).

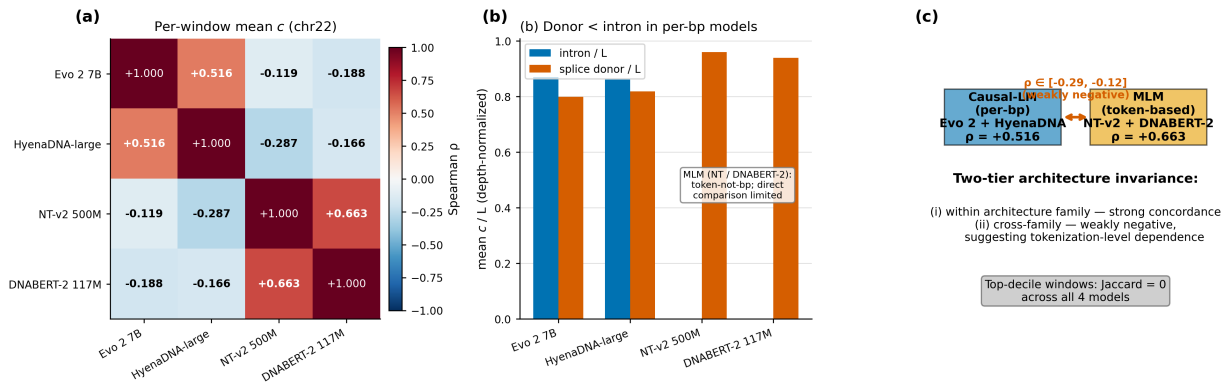


Figure 6. Cross-architecture two-tier structure (numerical detail in §3.3 body and Tables 5, 6). **(a)** Pairwise Spearman ρ of per-window mean settling depth across four genomic foundation models on chr22. **(b)** Depth-normalised donor vs intron mean c/L : per-bp causal-LMs (Evo 2, HyenaDNA) preserve donor < intron; tokenised MLMs (NT-v2, DNABERT-2) tokenise at window level and therefore show no per-position separation. **(c)** Schematic of the two-tier structure with the within/cross-family ρ values.